

In-Service Transceiver Calibration for Digital Subcarrier Signals with Synthesising IQ Imbalance via Offloaded Multi-layer Filters

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Abstract We proposed Tx/Rx-IQ imbalance calibration for digital-subcarrier signals using synthesised responses from multi-layer filters. In 1,200-km WDM transmission of 128-GBd 8-DSC PM-PCS-64QAM, Q-penalties within 0.1 dB under 2.5-ps Tx-IQ skew, 3-ps Rx-IQ skew, 2.5-dB Tx-IQ peaking error, and 63-GHz Rx-IQ imbalanced bandwidth limitation were demonstrated. ©2026 The Author(s)

Introduction

Growing traffic drives high-symbol-rate, high-information-rate (IR) digital coherent transmission [1], where frequency-dependent IQ imbalance has emerged as a significant limitation. Digital subcarrier (DSC) multiplexing offers improved nonlinear tolerance [2–3], enhanced equalization-enhanced phase noise robustness [4–5], flexible per-SC loading under bandwidth (BW) constraints, and resource-efficient implementations [5]. Nevertheless, SCs located at higher frequencies are particularly susceptible to component BW limitations and IQ imbalance, making impairment mitigation crucial. Adaptive equalizers have been proposed to mitigate these impairments [6–8]. However, implementing continuous adaptation in ASICs is often resource-intensive, while these impairments generally drift slowly and thus do not necessitate continuous adaptation.

Coherent DSP typically employs frequency-domain equalizers (FDEs) in the transmitter (Tx) and receiver (Rx) for factory calibration. These FDE coefficients can also be updated in service to track slowly varying IQ imbalance deviations, such as in-service device calibration [9–10]. We proposed an in-service Tx/Rx IQ imbalance calibration scheme based on an S/W offloaded multi-layer (ML) filter with extracting IQ imbalance in single-carrier signals where signal quality can be maintained without excessive cost [10]. For in-service calibration in DSC systems, full-band frequency-response calibration is desirable; however, the calibration must rely on pilot-aided per-SC compensation to avoid signal disruption. We

previously proposed adaptive ML filters with symmetric pairs to compensate Tx/Rx IQ distortion in DSC signals [8]. To extend this ML-based approach to enable in-service, full-band IQ-imbalance calibration for DSC, frequency-domain synthesis, which we previously investigated for strictly linear responses [11], is required for widely linear responses.

In this paper, we propose an in-service calibration scheme for Tx/Rx IQ imbalance in DSC signals. Tx/Rx IQ imbalance is extracted from per-SC filter coefficients of offloaded ML filters, synthesised into full-band Tx/Rx FDE coefficients, and applied in the adaptive main DSP. We experimentally demonstrated the effectiveness of the proposed method in a 128-Gbaud 8-DSC polarization multiplexing-probabilistic constellation shaping-64 quadrature amplitude modulation (PM-PCS-64QAM) WDM transmission over 1,200-km of single-mode fibre (SMF), where successful calibration of Tx/Rx IQ skew and frequency-response deviations was achieved.

In-service Tx/Rx calibration for DSC signals with synthesising IQ Imbalance via offloaded adaptive ML filters with symmetric pairs

Since Tx/Rx impairments vary slowly, the proposed in-service calibration uses S/W offloaded ML filters to derive Tx/Rx FDE coefficients, eliminating the need for dedicated hardware and preventing signal disruption. Fig. 1 illustrates the conceptual overview to generate synthesised Tx/Rx calibration filter coefficients. The adaptive ML filters with symmetric pairs in Fig. 1 (a) stack

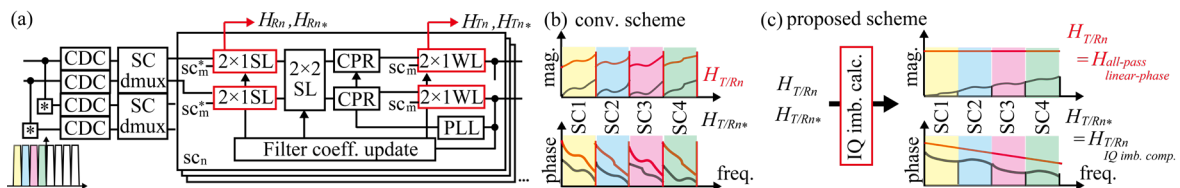


Fig. 1: Concept of IQ imbalance calibration filter coefficient generation for DSC signals (a) The block diagram of ML filters with symmetric pair, (b) Conventional filter coefficient frequency synthesis process across SC, (c) Proposed IQ imbalance compensation filter coefficient frequency synthesis process across SC.

strictly linear (SL) and widely linear (WL) layers, with filter coefficients updated via backpropagation from the final outputs. Chromatic-dispersion compensation (CDC) is first applied to both the received DSC signal and its complex conjugate, followed by SC demultiplexing. The ML chain performs Rx compensation with 2×1 SL filters, polarization demultiplexing with a 2×2 SL filter, carrier phase recovery (CPR), and Tx compensation with 2×1 WL filters per-SC basis. Unlike the single-carrier case, each DC-symmetric SC pair is processed jointly, as IQ imbalance generates conjugate-image interference between SCs mirrored around DC. Data-aided phase-locked loop (DA-PLL) using the pilot symbols is applied for CPR, and it is embedded for the least mean square (LMS) method to update filter taps.

We synthesise the per-SC coefficients obtained from the 2×1 WL/SL filters in the frequency domain to reconstruct the overall response across the full DSC BW. The conventional filter coefficient synthesis process, where we simply stitch the per-SC coefficients, is described in Fig. 1(b). Here, $H_{T/Rn}$, $H_{T/Rn}^*$, and n denote the frequency response of the 2×1 WL/SL filter coefficients, and SC index. Since $H_{T/Rn}$ and $H_{T/Rn}^*$ are optimized independently, simple stitching can produce magnitude/phase jumps at SC boundaries; additionally, SC dependent variations of the IQ common component can also introduce gain/phase offsets, resulting in residual compensation error and performance penalties.

The proposed IQ imbalance compensation filter coefficient synthesis process is described in Fig. 1(c). We transform the per-SC 2×1 WL/SL filter coefficients in the form only to compensate for IQ imbalance before stitching to align the SC-to-SC gain/phase reference and remove the SC-dependent IQ common component. The filter coefficient transformation for compensating only the Tx-IQ imbalance is given as follows:

$$Y_{Tn}(\omega) = X_{Tn}(\omega) - \frac{H_{Tn*}(\omega)}{H_{Tn}^*(-\omega)} X_{Tm}^*(-\omega), \quad (1)$$

where, Y_{Tn} , X_{Tn} , ω , and m denote the output signals, the input signals, angular frequency, and DC-symmetric SC pair index, respectively. Similarly, the filter-coefficient transformation for the Rx side is described as:

$$Y_{Rn}(\omega) = X_{Rn}(\omega) + \frac{H_{Rn*}(\omega)}{H_{Rn}(\omega)} X_{Rm}^*(-\omega), \quad (2)$$

where Y_{Rn} , X_{Rn} is the output signals and the input signals, respectively. In this form, the direct-path from $X_{T/Rn}$ to $Y_{T/Rn}$ becomes all-pass linear phase for all SCs and only the conjugate branch remains active [10], enabling discontinuity-free stitching and avoiding IQ common overcompensation. Any dips at these boundaries are filled using linear interpolation [11]. As prerequisites, CDC and signal normalization are applied to the DSC signal before SC demultiplexing, and a common filter length is used across all SCs to maintain a consistent magnitude and phase reference. Finally, the synthesised filter coefficients are converted into 2×2 real-valued IQ-MIMO coefficients for each polarization [9] and applied to the Tx/Rx FDEs in the main DSP for in-service calibration.

Experimental setup and offline DSP

Fig. 3 illustrates the experimental setup and offline DSP for 128-GBd 8-DSC PM-PCS-64QAM signals with an IR of 4.1 b/sbl/pol. Root-raised-cosine (RRC) filtering was employed with a roll-off factor of 0.015625. External cavity lasers (ECLs) served as the signal source and local oscillator (LO). On the Tx side, the electrical signal was generated by four 256-GSa/s DACs and fed into a coherent driver modulator (CDM) class 60 [12]. DVB-S2 LDPC FEC with a frame length of 64,800 and a code rate of 4/5 was applied. For each SC and polarization, eight FEC frames were formed by loading random payload bits and mapping them to PM-64QAM using PCS with constant-composition distribution matching (CCDM). Pilot-aided DSP was enabled by inserting one quadrature phase shift keying (QPSK) pilot every 25 symbols, together with an additional 2^{10} symbol QPSK overhead for filter pre-convergence [13]. The SCs were multiplexed in the frequency domain to generate the DSC signal. A pre-calibrated fixed Tx filter compensated for Tx device frequency response. X-IQ skew was digitally emulated, and the driver X-Q peaking inside the CDM was adjusted to induce frequency-dependent IQ imbalance.

For WDM transmission, the channel under test was surrounded by ten neighboring channels generated by amplified spontaneous emission (ASE) and a wavelength-selective switch (WSS) on a 150 GHz grid. The WDM signal was transmitted over 1,200-km, comprising twenty 60-km SMF spans with inline EDFAs. The launch power

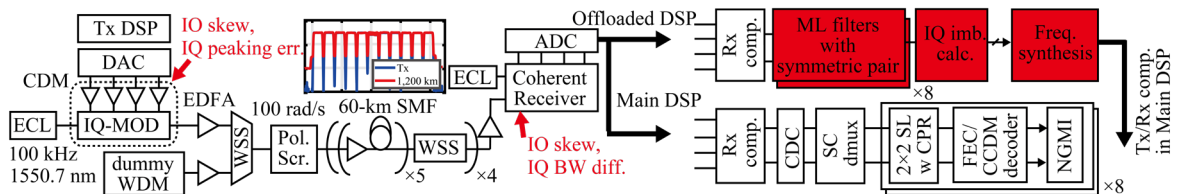


Fig. 2: Experimental setup and offline DSP

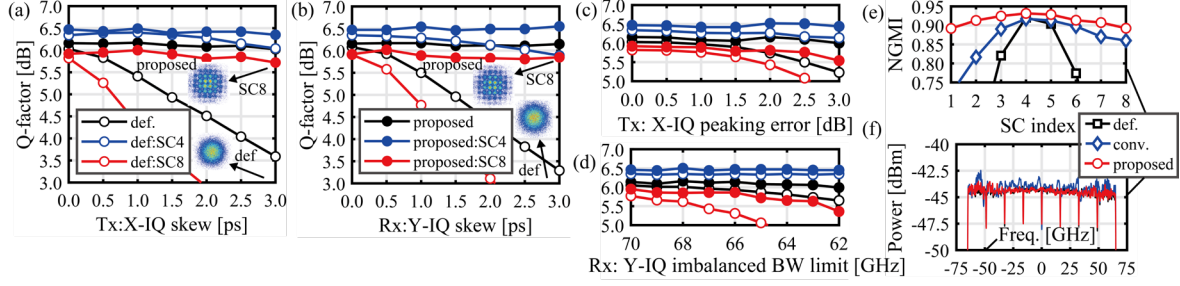


Fig. 3: Experimental results (a) Q-factor as a function of Tx X-IQ skew, (b) Rx Y-IQ skew, (c) Tx X-IQ peaking error of drivers inside CDM, (d) Rx Y-IQ imbalanced BW limitation, (e) NGMI of each SC with multiple impairments, (f) Tx optical spectrum with multiple impairments (res.: 500 MHz)

was set to +1 dBm per channel, resulting in a received OSNR of 29.0 dB/0.1 nm after 1,200-km. The optical signal was coherently received and digitized using four 256-GSa/s ADCs with a LO offset of approximately 100-MHz. Y-IQ skew and Y-IQ imbalanced BW limitation, implemented via a 5th-order Butterworth filter, were digitally emulated at the ADC outputs.

Validation used two independent data sets: one to derive the Tx/Rx FDE coefficients using the proposed frequency synthesis, and another to evaluate performance with Tx/Rx FDE-based calibration. A 2-Sa/symbol signal after Rx front-end compensation was processed by the ML filters with symmetric pairs (9 taps for 2×1 SL/WL, 33 taps for the 2×2 SL), updated via pilot-aided DA-LMS. The synthesised filter coefficients were subsequently applied in the main DSP for performance evaluation. For comparison, a conventional frequency synthesis scheme in Fig. 1(b) was also tested. The main DSP performed CDC and SC demultiplexing, followed by a 33-tap 2×2 SL filter with PLL-based CPR per SC. Q-factors were derived from the pre-FEC BER and the normalized generalized mutual information (NGMI) averaged over two polarizations and SCs.

Results and Discussion

First, we evaluated the proposed scheme under various IQ imbalance conditions and compared its performance to the baseline without Tx/Rx FDE-based calibration. The Q-factor is plotted as a function of Tx-X-IQ skew in Fig. 3 (a). To further examine SC-dependent differences, we also plotted the Q-factors for SC4, an inner SC, and SC8, an outer SC. Without calibration, the Q penalty increased with the SC carrier frequency (for 2.5 ps: average penalty 2.0 dB, 0.2 dB for SC4, and 3.6 dB for SC8). With the proposed scheme, the penalty was 0.1 dB at 2.5 ps and remained low even at 3 ps (0.2 dB for SC8). Fig. 3(b) presents the Q-factor as a function of Rx Y-IQ skew. A similar trend was observed, with the proposed calibration limiting the Q penalty to 0.1 dB up to 3 ps. Fig. 3(c) shows the Q-factor as a function of Tx

X-IQ peaking error. At 2.5 dB, the baseline penalties were 0.2 dB (SC4) and 0.7 dB (SC8), while the proposed scheme reduced them to 0.1 dB on average and 0.2 dB for SC8. Fig. 3(d) presents the Q-factor as a function of Y-IQ imbalanced BW limitation. With the proposed calibration, SC8 exhibited a 0.1 dB penalty at 66 GHz, rising to 0.6 dB at 62 GHz, whereas the baseline penalty reached 1.5 dB at 66 GHz since excessive BW limitation directly suppresses the signal BW of the outer SCs. Nevertheless, an average penalty was 0.1 dB at 63 GHz using the proposed method.

Finally, we compared the proposed scheme with the conventional method under combined impairments: 2 ps Tx X-IQ skew, 2 ps Rx Y-IQ skew, 1.5 dB Tx X-IQ peaking error, and 66 GHz Rx Y-IQ imbalanced BW limitation. In Fig. 3(e), NGMI is plotted as a function of SC index and Fig. 3(f) shows the Tx optical spectra. The conventional method suffered from inter-SC discontinuities and degradation due to overcompensation of IQ common distortion, which was most pronounced on the outer SCs; for example, SC1 exhibited an NGMI below 0.75. This tendency was also evident in the optical spectrum. In contrast, the proposed scheme mitigated most of the combined impairments and achieved NGMI exceeding 0.89 for all SCs, also no post-FEC errors were observed, with only minor residual degradation on the outer SCs, demonstrating the robustness of the proposed IQ imbalance synthesis scheme.

Conclusions

We have proposed an in-service transceiver calibration scheme for DSC signals that synthesises Tx/Rx IQ imbalance compensation using off-loaded ML filters with symmetric pairs. The scheme was experimentally validated in a 1 Tb/s 128-GBd 8-DSC PM-PCS-64QAM WDM transmission over 1,200-km of SMF, demonstrating robust calibration via synthesised IQ imbalance compensation. Negligible Q penalties below 0.1 dB were achieved under 2.5 ps Tx X-IQ skew, 2.5 dB Tx X-IQ peaking error, 3 ps Rx Y-IQ skew, and a 63 GHz Rx Y-IQ imbalanced BW limitation.

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